

LAB 02

Swing

Swing: Teacher-Assisted Acquisition and PPO Refinement

Environment design · PPO · Honest verification

BUILD → TRAIN → VERIFY → EXPLAIN



What you will build — and what the evidence says

- 01** Use case: Teach exploration gaps, privileged-teacher labeling, observation-only behavior cloning, DAgger, strict physical checkpoint selection, and cautious PPO refinement.
- 02** By the end: describe the scene, map observations to actions, explain the reward, trace one PPO update, run a bounded experiment, and defend an evaluation claim.
- 03** The large-amplitude behavior was acquired through privileged teacher labels, BC, and DAgger; it was not discovered by fresh PPO.
- 04** Scope: local MuJoCo simulation. Do not deploy these classroom artifacts to a physical robot; hardware transfer requires a separate validated process.

Materials and prerequisites · 1/2

- 01 Apple Silicon Mac with the installed RLX environment, MLX, MuJoCo, ONNX, ONNX Runtime, rendering dependencies, and Node.
- 02 Tracked teacher definition: docs/remaining-scenarios-e2e/recipes/swing-teacher.json.
- 03 Swing simulator assets: retained seat, strap, buckle, bumpers, frame, anchors, attachments, and two compliant tendons.
- 04 Bootstrap tool: rlx/scripts/bootstrap_swing_e2e.py; PPO/eval/export tool: rlx/examples/ppo_microduck_studio.py.

Materials and prerequisites · 2/2

05

Saved bootstrap provenance, demonstrations, selected initializer, PPO checkpoint, sidecar, ONNX actor, audit traces, and MP4.

Environment and policy contract · 1/2

- 01 Nominal XML MuJoCo swing at 50 Hz; accepted evaluation starts at angle 0 deg and rate 0 rad/s and runs 1,200 steps or 24 seconds.
- 02 Observation remains float32 batch x 61; no privileged swing angle or rate is appended to the deployed actor input.
- 03 Swing command slots are twist 48:51 = (0, body-y-axis-x, body-y-axis-z); head and body command blocks are zero.
- 04 The actor emits 14 values; the wrapper clips to [-1,1], projects to sagittal symmetry, and targets SEATED_POSE + 0.7 x projected action.

Environment and policy contract · 2/2

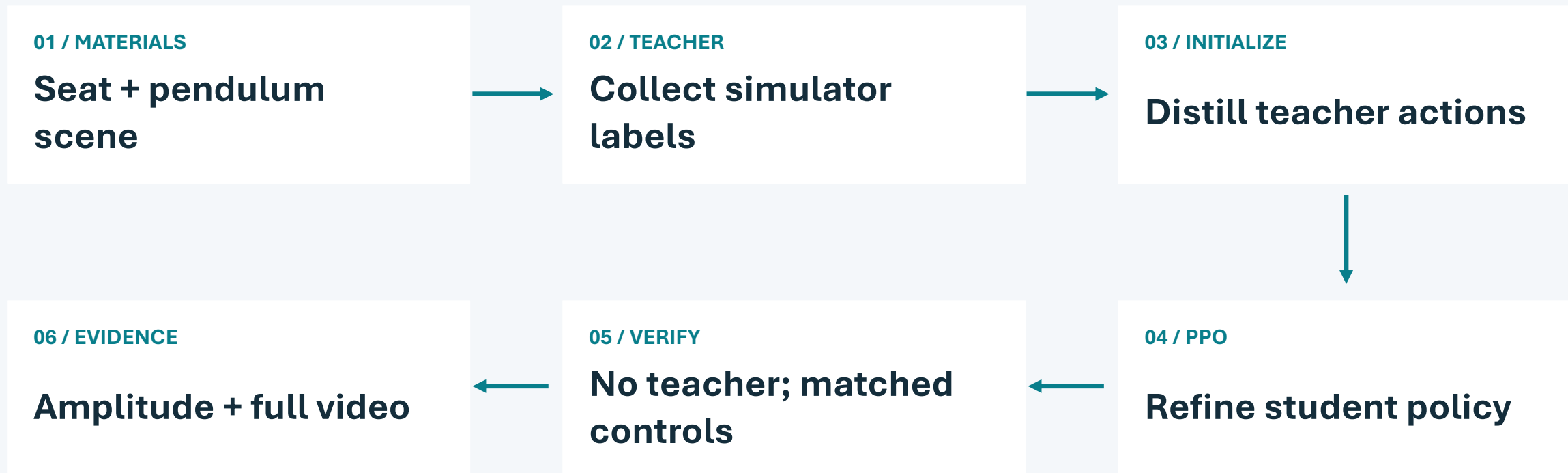
05

Yaw/roll action indices 0,1,7,8,9,10 are zeroed; leg pitch/knee/ankle pairs are equal and opposite; neck and head pitch remain independent.

06

The selected run disables domain randomization, observation noise, action delay, random yaw, and assisted evaluation starts.

How this case is built: one evidence loop



Failure is data: change one cause, retrain, and repeat the same evaluation gates.

Build the case step by step · 1/2

- 01 Use the bounded privileged teacher to collect eight 1,200-step episodes across four angle/rate reset bounds, yielding 9,600 labeled rows.
- 02 Freeze observation normalization from the initial 9,600 observations using sensor-block variance floors.
- 03 Train the actor mean with behavior cloning, then run DAgger with student actions and teacher labels on learner-visited states.
- 04 Evaluate every BC/DAgger stage physically; select stage 3 because it passes, not because it has the lowest supervised loss.

Build the case step by step · 2/2

05

Copy the selected checkpoint and sidecar into a fresh Studio run; retain the random critic and declare teacher assistance and ppo_steps 0.

06

Refine actor and critic with 524,288 PPO transitions, frozen observation normalization, fresh Adam state, and reward normalization.

07

Export deterministic ONNX, run strict still-start evaluation, audit trained/initial/zero/random controls, and validate the complete video.

PPO: improve the actor without trusting a huge step

$$\text{ratio} = \pi_{\text{new}}(a \mid o) / \pi_{\text{old}}(a \mid o)$$

Maximize $\text{mean}[\min(\text{ratio} \times A, \text{clip}(\text{ratio}, 1 - \epsilon, 1 + \epsilon) \times A)]$

Actor: chooses joint actions. Critic: estimates future discounted reward.

A = advantage. ϵ = clipping range. Old log probabilities stay fixed for each collected batch.

Follow one rollout through the learner

- 01 Collect observations, sampled actions, old log probabilities, values, rewards and episode-boundary flags using the current stochastic actor.
- 02 Compute TD residuals and generalized advantage estimates (GAE). A true terminal has zero continuation value; a timeout may bootstrap the saved final observation.
- 03 Cut advantage traces at reset boundaries. Never bootstrap a timeout from the first observation of the next episode.
- 04 Reuse this batch for PPO minibatch updates, then collect fresh on-policy data. Export the deterministic actor and its observation transform for evaluation.

Paper exercise: count samples and clip the objective

- 01 Illustrative arithmetic, NOT this run: $16 \text{ environments} \times 128 \text{ steps} = 2,048$ fresh transitions per rollout. Four epochs reuse those transitions; they do not create 8,192 new samples.
- 02 With $\epsilon = 0.2$, ratio = 1.4 and $A = +2$: $\min(2.8, 2.4) = 2.4$. The minimized policy term is -2.4 before averaging and additional loss terms.
- 03 With the same ratio and $A = -2$: $\min(-2.8, -2.4) = -2.8$. Explain why clipping is asymmetric once the sign of the advantage changes.
- 04 Run from repo root: `python3 docs/microduck-training-student-book/learning_lab.py --gae`. This is a deterministic arithmetic lab, not new policy training.

Reward design: what the agent is paid to do · 1/2

- 01 swing_peak_progress 224: pays bounded increments only when a new positive or negative angular frontier is reached.
- 02 swing_height 8, swing_late_height 24, and swing_energy 0.5 pay valid height and mechanical-energy gain above the reset baseline.
- 03 swing_lateral_penalty 3, lateral_barrier 8, lateral_velocity 3, and out_of_plane 1 penalize non-sagittal motion.
- 04 swing_alignment_penalty 18 and alignment_barrier 4 penalize seat/string geometry misalignment.

Reward design: what the agent is paid to do · 2/2

05

string_slack_penalty 8 and string_extension_penalty 12 protect the declared tendon-length region.

06

invalid_episode_penalty 10, action_rate_penalty 0.03, joint_torque_penalty 0.001, and joint_limit_penalty 1 are active.

Recorded training recipe · 1/2

- 01 Selected initializer: BC plus on-policy DAgger, stage 3, 38,400 teacher-labeled rows, strict pass, teacher_samples 38,400, ppo_steps 0.
- 02 Final demonstrations archive has 57,600 rows because collection continued after stage 3; it is not the selected checkpoint's training set.
- 03 PPO: seed 7, 524,288 transitions, 16 fork environments, 256 steps each, 4,096 transitions per rollout, and 128 rollout/update cycles.
- 04 Four minibatches x two epochs give 1,024-row minibatches, 8 optimizer steps per rollout, and 1,024 optimizer steps total.

Recorded training recipe · 2/2

05

Learning rate 0.000003; gamma 0.995; GAE lambda 0.98; clip 0.05; value coefficient 1; entropy coefficient 0; max gradient norm 0.5.

06

Initial standard deviation 0.1 is serialized through imitation; PPO uses a fresh Adam optimizer and freezes observation-statistic updates.

07

Update log fields are phase, steps, seconds, optimizer_steps, mean_loss, policy_loss, value_loss, entropy, approximate_kl, clip_fraction, explained_variance, env_steps.

Hands-on commands: inspect, smoke, train · 1/3

01

```
From .:  
./scripts/train-swing.sh --dry-run  
Inspect the complete teacher-initialization and PPO pipeline without training.
```

02

```
From .:  
./scripts/train-swing.sh  
Run the independent Full Swing bootstrap, PPO, export, evaluation, and render pipeline.
```

Hands-on commands: inspect, smoke, train · 2/3

03

```
From .:  
rlx/.venv-microduck/bin/python rlx/scripts/bootstrap_swing_e2e.py --source docs/remaining-  
scenarios-e2e/recipes/swing-teacher.json --output /tmp/swing-bootstrap-reproduction-  
YYYYMMDD-HHMMSS  
Create the teacher-assisted initializer; replace the timestamp token consistently.
```

04

```
From .:  
rlx/.venv-microduck/bin/python rlx/scripts/audit_scenarios.py --recipe-json docs/remaining-  
scenarios-e2e/recipes/swing.json --run rlx/runs/studio/swing/swing-reproduction-YYYYMMDD-  
HHMMSS --output rlx/artifacts/scenarios-e2e-20260907/swing-reproduction-YYYYMMDD-HHMMSS-  
audit --seeds 501 502 503 504 505 --render  
Audit a completed Swing reproduction over five deterministic seeds.
```

Hands-on commands: inspect, smoke, train · 3/3

05

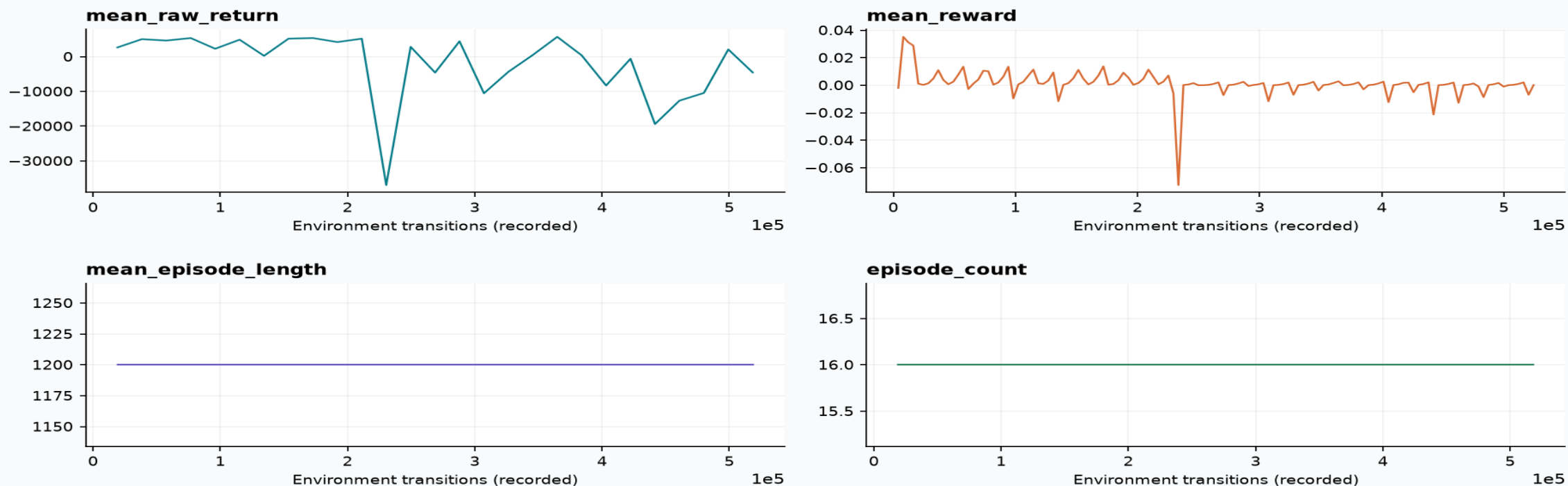
From .:

```
jq '{selected_stage, samples, normalizer_frozen_sample_count, limitations}' /tmp/swing-bootstrap-reproduction-YYYYMMDD-HHMMSS/provenance.json
```

Inspect selected bootstrap stage, sample count, frozen normalizer, and declared limits.

Reward: does the logged signal match the task?

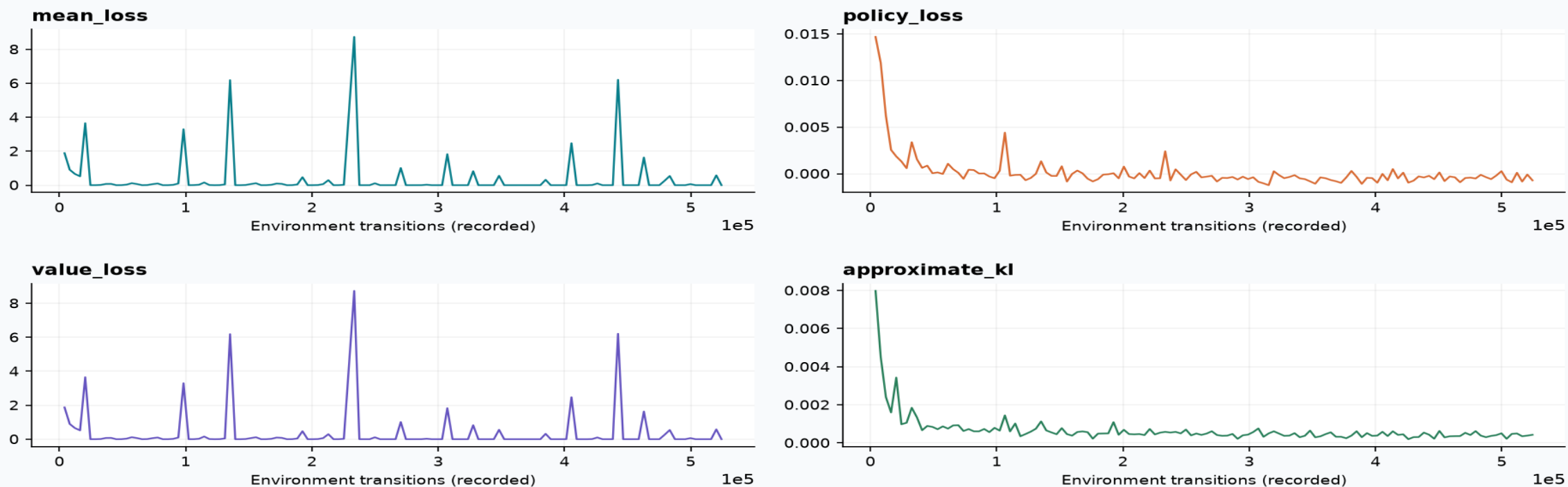
Recorded reward signals — different definitions, not interchangeable



Training reward is an optimization signal, not the acceptance test. See `raw/<case>/` for complete recorded fields.

Loss: inspect actor, critic and policy drift

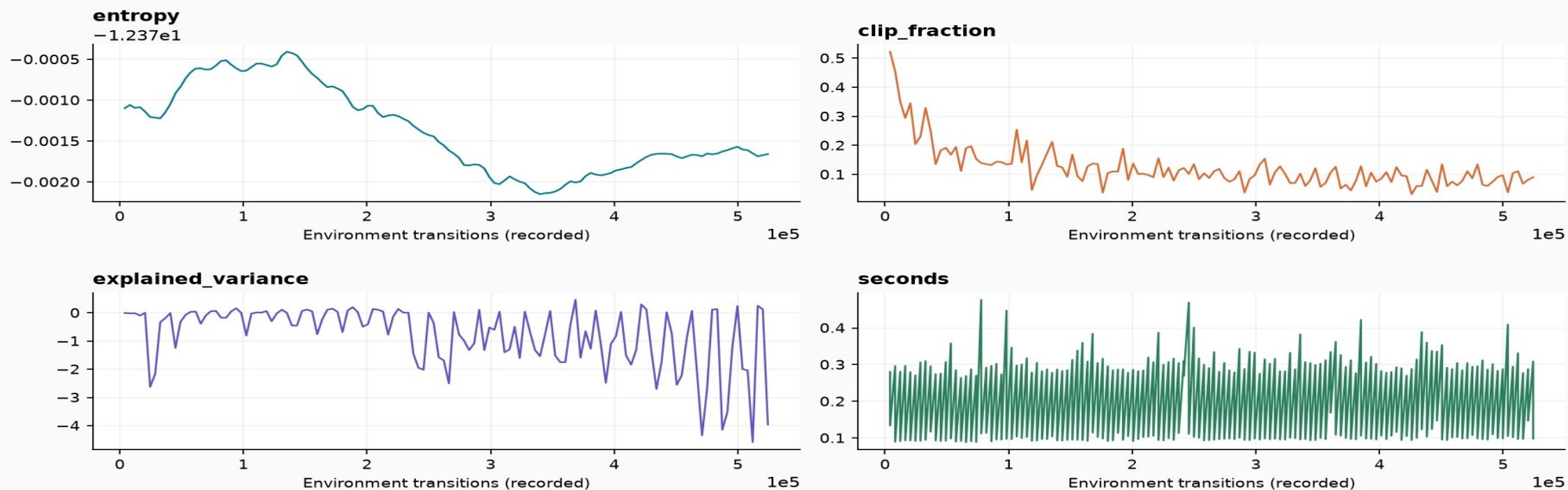
Recorded PPO optimization — raw values, no smoothing



A lower loss does not prove a better robot skill. Loss sign and scale depend on the implementation.

Diagnostics: is the optimizer behaving?

Recorded update diagnostics — absent fields are explicit



Entropy, clipping, explained variance and timing are diagnostics — not a standalone pass/fail certificate.

Metric matrix: numerical audit of this trace · 1/3

- 01 approximate_kl: first 0.007981 → final 0.0004294; min 0.0001979, max 0.007981; 128 logged samples.
- 02 clip_fraction: first 0.5221 → final 0.09058; min 0.03223, max 0.5221; 128 logged samples.
- 03 entropy: first -12.37 → final -12.37; min -12.37, max -12.37; 128 logged samples.
- 04 explained_variance: first -0.002574 → final -3.963; min -4.591, max 0.4628; 128 logged samples.

Metric matrix: numerical audit of this trace · 2/3

- 05 mean_loss: first 1.889 → final 0.0001821; min -0.0003026, max 8.735; 128 logged samples.
- 06 mean_raw_return: first 2670 → final -4533; min -3.695e+4, max 5725; 27 logged samples.
- 07 mean_reward: first -0.001981 → final 0.0001209; min -0.07257, max 0.03519; 128 logged samples.
- 08 policy_loss: first 0.01469 → final -0.0007164; min -0.001231, max 0.01469; 128 logged samples.

Metric matrix: numerical audit of this trace · 3/3

09

value_loss: first 1.874 → final 0.0008984; min 0.0003912, max 8.733; 128 logged samples.

Verification: define success before looking · 1/2

- 01 Require one complete 1,200-step still-start episode per lane and symmetric bidirectional span ≥ 150 deg within each episode.
- 02 Require every sample to have lateral offset ≤ 0.020 m, alignment ≤ 0.050 , string lengths 0.370-0.394 m, valid geometry, and positive tension.
- 03 Reject pooling a positive peak from one episode with a negative peak from another; all episodes must pass independently.
- 04 Confirm the exported actor accepts only 61 observations and emits 14 actions; privileged angle/rate access is confined to teacher labeling.

Verification: define success before looking · 2/2

05

Compare final PPO policy with selected initializer, zero action, and random action; require negative controls to fail the skill gate.

06

Validate checkpoint and sidecar hashes, JSONL finiteness, dynamic batch x 61 to x 14 ONNX, parity $\leq 1e-4$, decoded MP4, and browser playback.

Results: what was actually measured · 1/3

- 01 Actual final evidence: PASS for swing-e2e-20260907-v3; Studio evaluation passed 16/16 complete 24-second episodes.
- 02 Independent seeds 501-505 passed 5/5; nominal deterministic resets produced the same 162.3215866231874 deg bidirectional span.
- 03 Final peaks were +82.94345706612039 and -81.1607933115937 deg; max lateral offset 0.00901905817024827 m; max alignment 0.04606865972943153.
- 04 Valid geometry and both-strings-tensioned fractions were 1.0; string lengths stayed 0.38010451970068404-0.38667901638228347 m.

Results: what was actually measured · 2/3

- 05 Minimum sampled spring tension was 0.20903940136807186 N; deterministic raw return was 7016.894722654329.
- 06 The selected imitation initializer already passed at about 154.62 deg span and raw return 6126.39; PPO improved return by 890.5029066062816.
- 07 PPO changed learned tensors by L2 0.4897554898284089; ONNX/native maximum action error was 1.1324882507324219e-6.
- 08 Final-100 means: policy_loss -0.00032241872322629205, value_loss 0.37311747095896863, mean_loss 0.37279505291702664.

Results: what was actually measured · 3/3

09

Final-100 approximate_kl 0.00047223041314282455, clip_fraction 0.0969189453125, explained_variance -0.8998712115795674.

10

A scratch-PPO run reached only about 2.23 deg span; a 1,048,576-transition continuation kept large span but failed geometry and tension.

Watch the actual saved rollout



BEFORE PLAY

Predict one success signal and one failure signal.

AFTER PLAY

Which claims does this clip support? What remains untested?

24.0 s saved MP4 • Click in PowerPoint Slide Show • Backup: videos/swing-rollout.mp4

Limits and failure analysis · 1/2

- 01 The large-amplitude behavior was acquired through privileged teacher labels, BC, and DAgger; it was not discovered by fresh PPO.
- 02 The selected nominal seeds are nearly identical because randomization, noise, delay, yaw, and nonzero start bounds are disabled.
- 03 Positive sampled spring tension excludes safety-limit constraint forces and does not establish continuous-time or hardware load safety.
- 04 The critic remained poorly fitted by explained variance even though the deterministic actor passed the separate physical gate.

Limits and failure analysis · 2/2

05

The accepted result uses nominal XML simulation, one teacher family, one PPO seed, planar actions, and no disturbance or BAM evaluation.

06

Further PPO can damage a competent policy; the rejected million-transition checkpoint is not an acceptable alternative.

Student lab: change one variable, keep the gates · 1/2

- 01 Recommended exercise, not saved evidence: derive the 13-parameter teacher equation and map its five sagittal outputs into 14 actions.
- 02 Recommended exercise, not saved evidence: verify cumulative DAgger rows and optimizer updates for stages 0 through 5.
- 03 Recommended exercise, not saved evidence: tabulate actor, critic, log std, observation normalizer, return normalizer, and optimizer at handoff.
- 04 Recommended exercise, not saved evidence: derive 4,096 transitions per rollout, 128 rollouts, and 1,024 PPO optimizer minibatches.

Student lab: change one variable, keep the gates · 2/2

05

Recommended exercise, not saved evidence: explain why lower BC loss at stages 4 or 5 cannot override a failed physical gate.

Submit an experiment, not just a video

- 01** Before training: write one hypothesis, the changed parameter, the fixed baseline, seed list, acceptance criteria and time/transition budget.
- 02** During training: save the exact command, environment versions, initialization, checkpoint and raw reward/loss/KL metrics. Never overwrite the saved reference run.
- 03** After training: evaluate the frozen exported policy on matched seeds; include all failures, aggregate metrics, the full rollout video and a source-bound result table.
- 04** Rubric / 10: reproducibility 2; environment and reward explanation 2; PPO and metrics interpretation 2; honest verification 2; one evidence-based next experiment 2.

Source map and further study · 1/2

01

AGENTS.md

02

microduck_local/AGENTS.md

03

README.md

04

rlx/README.md

Source map and further study · 2/2

05

`docs/microduck-training-student-book/chapters/02a-ppo-code-lab.md`

06

`docs/microduck-training-student-book/chapters/02b-gae-workshop.md`

07

`scripts/scenarios/README.md`

08

`README.md`